A Training-Pool Replay Audit of the Released SLRTP2025 Back-Translation Evaluator: Non-Replication Across Independently Trained Checkpoints and a Readout-With-Generalization Signature

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Abstract

Back-translation (BT) evaluators mediate sign language production rankings, but a released checkpoint may yield orderings that do not survive retraining or reference replacement. On the 641-sequence PHX-public subset of SLRTP2025, a constructed whole-donor replay probe (TN-PURE-v1) scores 23.79 sacreBLEU under the released BT evaluator versus 12.78 for the recorded test poses, a gap of **+11.01** BLEU points (95% CI [**+9.61, +12.43**]). Across 43 independently trained checkpoints spanning dev BLEU-4 4.5–10.9 (all less competent than the released evaluator's 13.38), none reproduces a comparable gap (range [**−2.01, +0.25**]); 15 distillation students (dev BLEU-4 up to 10.9) show non-positive gaps (**−2.01** to **0.00**). The three $\boldsymbol{\alpha}$**=1.0** students produce empty or degenerate outputs (seeds 101 and 303: 641/641 empty; seed 202: 483/641 empty, 158 single-token "."), resulting in near-zero BLEU. This non-reproduction holds under a competence confound (the 14 same-recipe reconstruction seeds yield a Student-$\boldsymbol{t}$ prediction interval of [**−1.42, +0.32**] BLEU points), so the unobserved 10.9–13.4 interval could in principle harbor non-linear behavior; because Saunders et al. limited their rank-stability claim to similar model configurations, this audit is a stress case on a released checkpoint, not a strong counter-disproof of that claim. On the matched 461-item confidence=1 subset of Czehmann et al.'s human back-translations, replacing the speech-transcribed references attenuates the gap from **+10.95** to **+1.32** (88.0%, paired bootstrap CI for the attenuation [**7.92, 11.49**]); the full-641 attenuation of **89.8%** is reported as sensitivity analysis. The PURE decode loses a larger fraction of its score than GT across all metrics, so simple symmetric compression is insufficient, but this does not discriminate between reference-text defects and training-distribution alignment. The released evaluator also decodes its 7,060-item training pool at 78.8 BLEU (70.7% exact match) while retaining non-trivial in-domain dev/test performance

(13.38/12.78); pose–text permutation collapses train-pool BLEU from 78.6 to $\sim$1.0; an input-side test that shuffles pose tensors before decoding (holding IDs fixed) similarly collapses BLEU to $\sim$1.0, and output-equivariance verification confirms $\boldsymbol{E}(\text{pose}_{\boldsymbol{j}}, \text{slot}_{\boldsymbol{i}}) = \boldsymbol{E}(\text{pose}_{\boldsymbol{j}}, \text{slot}_{\boldsymbol{j}})$ for 100/100 items, ruling out a pose-ignorant pure ID/cache lookup; and test-split poses from the same signers score only 12.2 BLEU (3.2% EM). This post-hoc operational signature is not exhibited by any constructible reconstruction or distilled student. The evidence is consistent with the anomaly being a property of the released checkpoint and its (partly undocumented) data pipeline as far as observable on PHX-public; the specific mechanism is not identifiable from public artifacts.

Keywords: sign language production, back-translation evaluation, retrieval-augmented generation, multimedia quality assessment, evaluator non-replication

1 Introduction

Sign language production (SLP) systems are commonly evaluated by back-translation (BT): a trained sign recognizer decodes each output pose sequence, and a text metric compares the decoding with the reference. This measures recognizer-readable content through a learned measurement model, so system rankings may reflect both the evaluated outputs and the inductive biases of a particular evaluator checkpoint. When proposing BT evaluation for SLP, Saunders et al. Saunders et al. (2021) argued that changing the BT model changes absolute scores but preserves relative rankings and recommended BT for comparing between similar model configurations. The present audit tests a stress case of that claim, but because all constructible checkpoints in our family are less competent than the released evaluator (dev BLEU-4 4.5–10.9 vs. 13.38), our non-reproduction result is scoped to the public training recipe and does not constitute a strong test of their rank-stability proposition.

Prior work shows that BT scores can be modest for recorded real signing and can improve without faithful target conditioning Walsh et al. (2024); Hong and Košecká, Jana (2026), and sign-native or pose-aware metrics provide complementary evidence Kim et al. (2024); Imai (2025); Jiang et al. (2026). Nevertheless, BT remains central to benchmarks such as SLRTP2025 Walsh et al. (2025). An unresolved question is whether a retrieval–reference stress-test ordering persists across independently trained instances of the same BT architecture.

This question is exposed by a retrieval–reference reversal on PHX-public: under the original SLRTP2025 evaluator, source-conditioned whole-donor retrieval scores 23.79 sacreBLEU (0–100 scale), whereas the recorded test poses score 12.78. This neither establishes better signing by retrieval nor invalidates BT evaluation in general — it asks whether the reversal persists across evaluator checkpoints or depends on properties of the original evaluator.

We address this question in four stages. First, we compare the reversal with reference-pose, generative, and random-donor controls under the original evaluator, verifying protocol details (scorer equivalence, frame-rate handling, decoding parameters). Second, we evaluate the same fixed outputs with fourteen independently trained

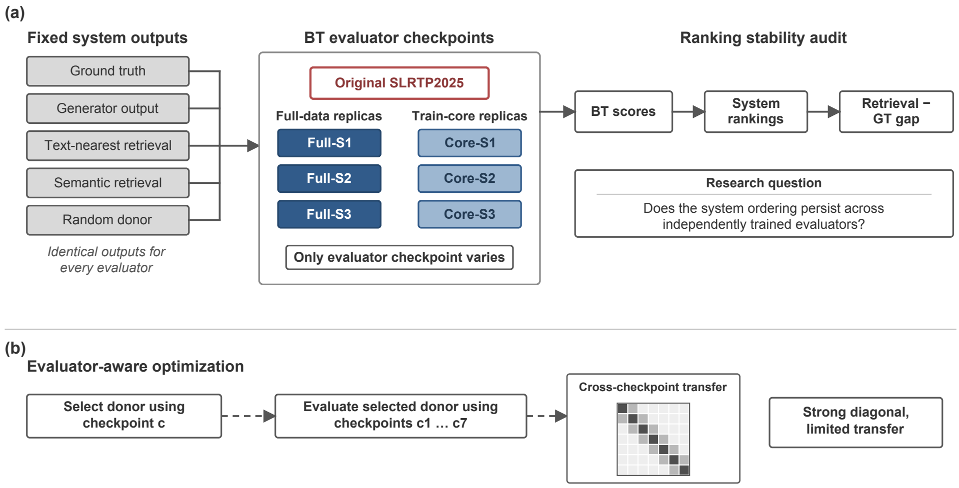

Fig. 1: Overview of the multi-checkpoint evaluation audit. (a) A fixed set of reference-pose, generative, retrieval, and random-donor outputs is evaluated by the original SLRTP2025 BT checkpoint and fourteen independently trained recipe-conditioned reconstructions. (b) Evaluator-aware analysis: donors selected by maximizing likelihood under one checkpoint are evaluated by all seven checkpoints.

recipe-conditioned reconstructions and quantify paired uncertainty, documenting that no rescue variant, checkpoint-selection objective (NLL, decoded BLEU, decoded WER), or step-faithful run reaches competence parity. Third, we test reference validity: on the matched 461-item confidence=1 subset of public human sign-to-text back-translations Czehmann et al. (2026), the reversal attenuates from +10.95 to +1.32 BLEU points (88.0%). Fourth, we characterize what distinguishes the released pipeline via five diagnostic views of a training-content readout property. Evaluator-aware oracle analyses are in SI Sup. H because their unmasked advantage does not survive reference-overlap exclusion (Fig. 1).

Why a constructed probe matters.

Retrieval-augmented SLP entries draw donors from the evaluator's training pool, so the TN-PURE-v1 stress test isolates a risk that production systems face in diluted form. Methodologically, the audit contributes a template — provenance table, checkpoint registry, evaluator-replication controls — that other benchmarks can adopt for rank-stability testing.

Evidence-level guide.

Findings are classified by evidence level: (i) *direct observations* reproducible from public artifacts (PURE 23.79 vs GT 12.78; reversal is decoded-text only, with GT NLL 3.06 vs PURE 3.48); (ii) *family-side negatives* that are competence-confounded (43 checkpoints, gaps $[-2.01, +0.25]$, all at dev BLEU-4 $<$ 13.38; 14 same-recipe

PI $[-1.42, +0.32]$); (iii) *matched-subset reference sensitivity* (human-ref substitution on 461 confidence=1 items attenuates gap by 88.0%, consistent with Czehmann et al.'s prior finding); (iv) *post-hoc mechanism hints with input-side controls* (readout-with-generalization: 78.8 BLEU / 70.7% EM train vs 13.38 dev; code-path audit excludes direct target-token teacher forcing; input-side pose permutation with output-equivariance verification rules out a pose-ignorant pure ID/cache lookup under the tested decoding path; SI Sup. N); and (v) the specific training mechanism, which is unidentifiable without organizer disclosure. Throughout, **prespecified** analyses (marked C in SI Sup. C) are separated from **post-hoc** analyses (marked D); the headline gap was observed exploratorily and the estimand was frozen afterwards.

2 Related Work

Learned evaluation in sign language production.

Neural SLP maps spoken language or glosses to pose and often evaluates with a frozen sign recognizer Camgoz et al. (2018); Saunders et al. (2020). BT was proposed as an SLP evaluation protocol by Saunders et al. Saunders et al. (2021), who argued the choice of BT model shifts absolute scores but not relative rankings; our audit is a direct stress test of that rank-stability claim. SignBLEU Kim et al. (2024), SiLVERScore Imai (2025), and pose-native measures offer complementary views, but BT BLEU remains central to benchmarks such as SLRTP2025 Walsh et al. (2025).

Retrieval and shortcut risks.

Retrieval-based SLP ranges from phonetic composition to retrieval-enhanced generation Walsh et al. (2024,?); Zuo et al. (2025); Tasyurek et al. (2025). The SLRTP2025 report describes a leading retrieval-based entry, making donor copying pertinent Walsh et al. (2025). More generally, memorization in retrieval-augmented language modeling shows that strong task scores can reflect reproduction of training examples Carlini et al. (2021); Kandpal et al. (2022); Khandelwal et al. (2020). A shortcut observed under one learned evaluator does not show that the resulting ordering persists under another checkpoint.

Robustness of learned metrics and corpus audits.

Hong and Košecká Hong and Košecká, Jana (2026) show BT-BLEU can diverge from target faithfulness; Walsh et al. Walsh et al. (2024) document a low ground-truth BT baseline; Jiang et al. Jiang et al. (2026) recommend BT likelihood over decoded BLEU. Yazdani et al. Yazdani et al. (2026), BackTranslation2.0 Cory et al. (2026), and He et al. He et al. (2024) expose further fragilities of model-based generation metrics. Two 2026 PHOENIX-2014T audits bear directly on our reversal: Czehmann et al. Czehmann et al. (2026) released human back-translations and documented reference information loss; Alkain et al. Alkain et al. (2026) showed PHOENIX-2014T has the highest train–test text overlap among tested SLT corpora. Artiaga et al. Artiaga et al. (2025) showed signer-dependent evaluation overestimates SLT capability. None of these studies runs multi-checkpoint evaluator experiments or donor-pool substitution controls.

Table 1: Three-way data provenance for PHX-public. The released evaluator's training manifest matches the released pose materialization (7,060/515/641), not the full official PHOENIX-2014T split. All 41 missing IDs share the reason `absent_from_slrtp2025_released_pose_records` (SI Sup. A). SHA-256 hashes verify file integrity.

Split source	Train	Dev	Test
Official PHOENIX-2014T	7,096	519	642
Released SLRTP2025 pose (`.pt`)	7,060	515	641
Released evaluator `config.yaml` manifest	7,060	515	641
Missing from official	36	4	1

Table 2: Experimental protocol summary.

Setting	Recognizer	Layout	Split size	N eval	α
PHX-public	SLRTP2025 evaluator	178-joint	641 seq	641	0.88 (tuned)
PHX-holdout-MSKA	mska_slr (phoenix-2014t)	133-joint	1,538 win	1,538	0.50 (frozen)
CSL-Daily	mska (csl-daily)	133-joint	256 win	256	0.50 (frozen)

3 Methods

3.1 Evaluation design

Let E_c denote a BT evaluator checkpoint and $o_{i,n}$ the pose sequence for output condition i and item n. Checkpoint c produces $h_{c,i,n} = E_c(o_{i,n})$ and $B_{c,i} = M(\{(h_{c,i,n}, y_n)\}_{n=1}^{N})$, where y_n is the reference text and M is the corpus-level sacreBLEU functional unless stated otherwise. The values $B_{c,i}$ induce a ranking over fixed output conditions. We call a result *evaluator-pipeline-sensitive* when a substantive pairwise ordering changes across independently trained checkpoints under a fixed output set. The fixed primary set comprises recorded reference poses (GT), random donor copy, source-conditioned whole-donor retrieval, a Progressive Transformer generator, and retrieval–generator compositions; oracle-gloss variants are diagnostic controls.

The main evaluation uses PHX-public, a German weather-domain text-to-pose benchmark. The released SLRTP2025 pose records contain 7,060/515/641 sequences; one official test ID is absent from the released materialization, so every result is conditioned on the complete 641-record release (SI Sup. A). Table 1 gives the three-way provenance: the official PHOENIX-2014T split, the released SLRTP2025 pose materialization, and the released evaluator's actual training manifest (confirmed via `config.yaml` which references the same `train.pickle/dev.pickle/test.pickle` that ship as `train.pt/dev.pt/test.pt`). Supplementary analyses use **PHX-holdout-MSKA** (133-joint, 1,538 windows) and CSL-Daily (Mode B, 1,176 sequences; SI Sup. M). Table 2 summarizes the protocols.

All pose sequences are evaluated at 12.5 fps, matching the SLRTP2025 evaluator training distribution (`skeleton_subsample=2`). Every canonical pose bundle is *stored* at 12.5 fps after exactly one construction-time `[::2]` transform; the audit bypasses the released wrapper and calls the BT model directly, so no double subsampling occurs. A three-path regression confirms this: canonical and explicit 25→12.5-fps paths give identical GT BLEU-4 (12.78) and WER (85.77), while an erroneous second subsample degrades them (6.65 / 86.85; full table in SI). The released BT model applies *no* internal subsampling: its `subsample` attribute is set but never used. The official Codabench starter-bundle PT scores 1.59 BLEU under the released evaluator, matching the official 1.5942. PHX-public uses α=0.88 (selected on dev); PHX-holdout-MSKA and CSL-Daily use α=0.5.

3.2 Systems and retrieval controls

For each target, source-conditioned whole-donor retrieval scans the complete training pool after Unicode NFKC normalization, whitespace normalization and lowercasing, and selects the sequence with maximum source-text token-set Jaccard similarity. Oracle-gloss retrieval instead uses target gloss and is treated as an input-leakage diagnostic; random retrieval selects a donor uniformly. Donor exclusion removes exact normalized text or gloss duplicates or donors above threshold τ. Gloss-based exclusion uses target gloss and is therefore an offline diagnostic rather than a deployable inference procedure.

Text-nearest tie-breaking.

Jaccard ties are resolved by minimum character-level Levenshtein distance and then by minimum SHA-256 hash of the training key. The §3.2 algorithm also excludes donors whose NFKC-normalized text equals the query's (exact-normalized-text exclusion).

Two materializations. The headline observations throughout this paper (PURE 23.79, gap +11.01; Tables 4, 3, 5, 7) use an earlier frozen materialization that predates the exclusion fix: for 30 test items it selected donors with NFKC-normalized text identical to the query rather than excluding them. The 29-checkpoint re-decode panel and the distillation ladder (Table 12) use the full §3.2 algorithm with exclusion, yielding PURE 23.02 (gap +10.24). The donor-registry consistency note (§4.4) quantifies the impact: 610/641 items produce identical output; the 31 differing items are consistent with the exclusion rule explaining most differences. Both materializations are deterministic; the §3.2 algorithm is reproduced in the artifact at `scripts/build_canonical_panel.py`.

Retrieval composition.

TN-PURE-v1 uses whole-donor copy with [::2] subsampling to 12.5 fps, preserving the donor's original duration. TN-PTCOMP-v1 (a secondary control) composes a PT scaffold with donor hands and face via α-blending of upper-body joints (α=0.88 for PHX-public; full joint-set definitions and duration/word-count matching statistics in the artifact). The separately matched seen–unseen factorial retains 605/641 targets matched on duration and word count.

Donor-origin contrast retrieval systems.

To test whether the retrieval advantage co-occurs with donor origin, we construct frozen probe systems from the same 641 queries: **UNSEEN-PURE-v1** retrieves from the 640 other test poses (excluding the query itself, its sequence root, and exact-text matches); **CROSSFIT-PURE-A/B-v1** split queries by hash into 331/310 folds, each retrieving from the other fold; **SEEN-PURE-MATCHED-v1** selects a train-pool donor within ± 0.1 Jaccard of each UNSEEN donor (622/641 admit one); **SEEN-RAND640-MATCHED-v1** applies matched selection inside frozen 640-donor sub-pools. SEEN-PURE-v1 re-uses the canonical TN-PURE-v1 registry. Full construction rules (calipers, tie-breaks, signer constraints) are in the artifact; these are constructed probes, not deployable generators.

3.3 Evaluator checkpoint family

The primary inferential checkpoint family comprises the original frozen SLRTP2025 evaluator and six *recipe-conditioned reconstruction* runs trained from scratch with the documented architecture and a common frozen train/development partition (primary seeds 101–606; eight extension seeds 707–1405 use the identical protocol). The original bundle's `validations.txt` logs 202 validation points over 2,828 optimizer steps ($\approx$102 epochs), with decoded-BLEU selection reaching best dev BLEU-4 13.38 at step 1,820; intermediate checkpoints are not published. We use two reconstruction protocols: *paper-derived* (14 runs, legacy script, 300 epochs, NLL selection) and *released-config* (4 runs, up to 3,000 epochs, decoded-BLEU early stopping). Both share the documented architecture (3L/256/512ff, 8 heads), tokenizer, manifests (7,060/515), [:: 2] subsampling, Adam (1e-3, betas .9/.998, wd .001), batch size 256, and plateau $\times 0.8$ LR schedule (full provenance in SI).

Readout-with-generalization signature (post-hoc operational definition).

A checkpoint exhibits the readout-with-generalization signature if it has both (i) training-pool free-decode BLEU ≥ 50 *or* exact-match $\geq 50\%$, and (ii) dev BLEU-4 ≥ 8.0. These thresholds were defined after observing the released evaluator (78.8 BLEU / 70.7% EM, dev 13.38) vs. the reconstruction family (max 11.5 train / 9.9 dev, no simultaneous satisfaction). The thresholds are operational heuristics for this case study, not validated criteria.

3.4 Metrics and uncertainty

The recorded test poses (GT) are decoded by each BT recognizer and compared with the original reference text. We report corpus BLEU-4 on the 0–100 sacreBLEU scale as the primary metric Post (2018); sentence-level statistics (pass-through, slot-F1) are reported separately on the 0–1 scale. BLEU-1, chrF, ROUGE-L, BERTScore Zhang et al. (2020) (`bert-base-multilingual-cased`, no baseline rescaling, F1), BLEURT Sellam et al. (2020) (`Elron/bleurt-large-512`, a PyTorch-native substitute for BLEURT-20 with $r = 0.87$ rank correlation and -0.47 bias against canonical

BLEURT-20 on 100 German pairs), and WER (jiwer 3.1.0 normalized) provide complementary decoded-text controls; BT likelihood and pose-DTWp provide non-decoded controls.

Scorer equivalence.

The vendored official scorer (SacreBLEU 1.4.2) and the paper scorer (SacreBLEU 2.5.1, `tok:13a`, `smooth:exp`) give identical corpus BLEU on all 28 evaluation cells (maximum $|\Delta| = 0.0000$; byte-identical hypotheses, all n-gram orders non-zero). We report the 2.5.1/13a signature as primary.

We report two forms of uncertainty. *Item-sampling uncertainty* uses paired sequence-level bootstrap (10,000 resamples, seed 42, percentile 95% CIs): per item, n-gram sufficient statistics are precomputed once, and every resample re-aggregates them into a corpus BLEU. A pairwise difference is called detectable when its interval excludes zero. Significance testing of the primary estimand uses a paired approximate-randomization permutation; because the headline gap was first observed in exploratory analysis and the estimand was frozen afterwards, we report a descriptive Monte-Carlo estimate: 0 of 10,000 permutations exceed the observed gap ($p \leq 10^{-4}$, exact 95% binomial interval $[0, 0.00037]$). *Training-replicate uncertainty* is reported only for the recipe-conditioned population: a descriptive seed-level 95% Student-t prediction interval (not extrapolable to the released evaluator). The eight extension seeds are post-hoc additions and enter only that interval, never the six primary comparisons. The single primary estimand is the PURE–GT decoded-sacreBLEU gap under the released evaluator; all other contrasts are secondary and are Holm-corrected within a defined family or explicitly labelled descriptive (SI Sup. C).

4 Results

4.1 Retrieval reverses the reference ranking under the original evaluator

Under the original SLRTP2025 evaluator, source-conditioned whole-donor retrieval scored 23.79 sacreBLEU (0–100 scale), compared with 12.78 for the recorded test poses — a retrieval–ground-truth gap of +11.01 BLEU points (95% CI $[+9.61, +12.43]$; 10,000 sequence-level resamples). This fixed-checkpoint reversal motivates the analysis but does not indicate superior sign production.

Controls localized the observation to source-conditioned donor retrieval (Table 4): pure donor copy exceeded the oracle-gloss composed variant (11.8), canonical PT-composed (18.86), PT baseline (1.59), and 20-seed random-donor baselines (0.90 PURE, 0.77 COMP). Oracle-gloss retrieval (which uses reference gloss and is leakage) did not exceed the recorded test poses (11.4–11.8 vs 12.78). Token-normalized teacher-forced NLL (lower is better) does not reproduce the reversal: GT 3.060 vs PURE 3.479, COMP 3.411, PT 4.074. Likewise the normalized pose-DTW diagnostic does not validate semantics (GT 0 by construction vs PT/PURE/random 0.00678–0.00830). The reversal is visible under complementary text metrics on the same hypotheses (Table 3): sacreBLEU-4 +11.01, BLEU-1 +21.03, chrF +13.23, ROUGE-L +17.87, WER −13.90

Table 3: Decoded-text metric sensitivity under the original SLRTP2025 evaluator (641 sequences; beam 3, length penalty −1, max length 30) under the official implementations (§4.1). The reversal direction is consistent across decoded-text metrics; teacher-forced NLL and pose-DTWp do not reproduce it.

System	BLEU-4	BLEU-1	chrF	ROUGE-L	WER	BERTSc.
GT-v1 (reference poses)	12.78	34.40	34.59	35.20	85.77	0.757
PT-v1	1.59	16.91	19.03	17.26	101.68	0.669
TN-PTCOMP	18.86	46.35	41.88	45.17	77.36	0.793
TN-PURE	**23.79**	**55.43**	**47.82**	**53.07**	**71.87**	**0.822**
Gap PURE−GT	+11.01	+21.03	+13.23	+17.87	−13.90	+0.066

Table 4: PHX-public conditioning and construction controls (641 sequences; 12.5 fps). Columns distinguish whole-donor copy from PT-scaffold composition (α=0.88); all rows share the query set, donor pool, evaluator, and sacreBLEU protocol. Semantic retrievers (LaBSE 19.2/8.4, multi-SBERT 16.5/8.1) lie between the canonical retriever and oracle gloss; full rows in the artifact.

Conditioning	Pure donor copy	PT+retrieval composed
Source text (German Jaccard; original frozen materialization)	**23.79**	18.86
Oracle gloss (input leakage)	11.4	11.8
Random donor (seeds 0–19)	0.90 (SD 0.24)	0.77 (SD 0.19)

(better), BERTScore-F1 +0.066, BLEURT +0.165. We describe the observation as a *decoded-text-metric reversal* and use sacreBLEU-4 as the primary decoded metric.

Metric implementations.

All secondary metrics use the official SLRTP2025 implementations: they reproduce the official repository values (BLEU-4 12.777, chrF 34.585, WER 85.775, ROUGE 35.196 Walsh et al. (2025)) exactly. Implementation choices matter for chrF (corpus-level sacrebleu `CHRF` defaults; a sentence-level variant yields 48.05 instead of 34.59) and WER (jiwer 3.1.0 with official normalization; raw whitespace-split yields 79.26 instead of 85.78). BLEU-4 is invariant; we fix the official implementations throughout (ablations in SI).

Decoding-parameter sensitivity.

Re-decoding across beam size $\{1, 3, 5\}$, length penalty $\{-1, 0\}$, and max length $\{30, 50\}$ shows the reversal is not an artifact of decoding: the PURE–GT gap is +9.8 (greedy), +11.0 (beam 3), +11.6 (beam 5), with length penalty and max length negligible.

Pure donor copy outperformed the composed variant for every source-conditioned retriever, and source-text Jaccard produced the highest pure-donor score (23.79, ahead of LaBSE 19.2 and multi-SBERT 16.5), while oracle-gloss retrieval did not exceed the recorded test poses in either construction (11.4–11.8 vs 12.78): the reversal does not require oracle-gloss input leakage, and pure copy — not composition — is its strongest form.

4.2 Independent evaluator reconstructions do not reproduce the reversal magnitude

TN-PURE-v1 yields an original-evaluator gap of +11.01 BLEU points (95% CI $[+9.61, +12.43]$). Across fourteen reconstruction runs, gaps range from −1.10 to +0.13; twelve are negative, two are near-zero positive (seeds 101, 606). TN-PTCOMP-v1 yields an original gap of +6.09 $[+4.76, +7.41]$; all fourteen reconstruction estimates are negative. The seed-level 95% prediction interval is $[-1.4, +0.3]$ (mean −0.55, SD 0.39, 2/14 positive). For the six primary runs, original-minus-run difference-in-differences estimates range from +10.89 to +11.99; every 95% interval excludes zero.

4.2.1 Recipe-conditioned evaluator reconstructions

All fourteen reconstructions share frozen manifests, architecture, hyperparameters, and selection rule; seeds 101–1405 vary initialization and data order (Table 5).

Table 5: Canonical fixed-output evaluation on the released 641-pose PHX-public subset. PURE is TN-PURE-v1; COMP is TN-PTCOMP-v1. Gaps are relative to local GT (BLEU points). The six primary seeds carry precomputed paired-bootstrap intervals; the eight extension seeds (bottom row, ranges) reproduce the same near-zero or negative pattern post hoc. Per-cell values in the artifact.

Evaluator	GT	PURE	Gap	COMP	Gap
Original	12.78	23.79	**+11.01**	18.86	+6.09
Seed 101	8.57	8.65	+0.08	7.92	−0.65
Seed 202	8.81	7.83	−0.98	7.54	−1.28
Seed 303	9.67	8.96	−0.71	8.71	−0.96
Seed 404	7.19	6.83	−0.36	6.38	−0.81
Seed 505	9.05	8.16	−0.89	7.97	−1.08
Seed 606	8.09	8.22	+0.13	7.93	−0.16
Extension 707–1405 (range)	7.71–9.56	6.84–9.09	−1.10 to −0.19	6.90–8.83	−1.03 to −0.44

Competence is assessed on the released 515-sequence development subset, never on PHX-public for checkpoint selection. Original obtains dev BLEU-4 13.38 and WER 83.37 (official jiwer implementation, 0–100 scale); reconstruction runs obtain 6.69–9.07 and WER 88.21–94.83. Their selected validation NLL is 2.641–2.712 versus Original teacher-forced dev NLL 3.235, showing that likelihood and decoded recognition competence are not interchangeable. On PHX-public, reconstruction GT scores (7.19–9.67 BLEU points) likewise remain below Original 12.78. Consequently, evaluator competence and unavailable original-training details remain confounded with evaluator identity; the experiment identifies evaluator-pipeline sensitivity, not a causal checkpoint-identity effect.

Competence gate and trajectory analysis.

A checkpoint is competence-matched only if both $|\Delta\text{dev BLEU-4}| \leq 1.0$ and $|\Delta\text{dev WER}| \leq 3.0$ (the released evaluator's dev values are 13.38 / 83.37). These are prespecified operational thresholds; we do not claim they are domain-standard. None of the 52 non-distillation trained checkpoints passes either half: every checkpoint fails both (dev BLEU-4 falls short by 3.4–8.9; dev WER exceeds the 86.37 bound by 0.08–13.3). Along dense-checkpoint trajectories (170 saved epochs across eight seeds), no saved epoch reaches the ± 1.0 BLEU band (max 9.73) and the best WER (86.41) still exceeds the 86.37 bound.

The multi-objective Pareto frontier over every saved epoch under three selection objectives (dev NLL, decoded BLEU, decoded WER) also fails to reach the gate. We expanded the rescue search beyond learning rate: batch size $\{128, 512\}$, dropout $\{0.2\}$, weight decay $\{0\}$, label smoothing $\{0.1\}$, and a longer-budget variant (600 epochs, lr 5×10^{-4}), two seeds each (Table 6). All twelve new runs fail the gate (dev BLEU-4 6.8–9.9); the best (weight-decay 0, seed 202, dev BLEU-4 9.92) yields PHX-public GT 10.73 / PURE 10.98 (gap +0.25), extending the dose–response toward the gate without any reversal reappearing. The gate conclusion is insensitive to threshold: across a $\pm[0.5, 2.0]$ BLEU $\times$ $\pm[1.5, 4.5]$ WER grid (12 combinations), 0 of 52 trained checkpoints passes any. We frame the result as released-evaluator pipeline non-reproduction rather than checkpoint identity.

What the released training log establishes.

The bundle's `validations.txt` logs dev BLEU-4 reaching 10.01 at step 770, 12.2 at step 1,134, and 13.38 at step 1,820 (Fig. 2). All constructible reconstructions reach 8–10 by epoch 50–100 and plateau, while the released evaluator continues improving past epoch 30–60. The training dynamic that produced this slope is not explained by public artifacts.

Competence matching by input degradation.

Spatial jitter ($\sigma \leq 8\%$) barely moves either score; temporal downsampling steadily compresses both. At k=8 (one frame in eight), GT reaches 8.75 (inside the reconstruction range), yet PURE still scores 11.69 (gap +2.94). The replay advantage attenuates with input quality but is not eliminated, bounding the competence component without identifying it, since degradation also constitutes a distributional shift. The full degradation table (8 rows) is in SI Sup. F.

The six primary seeds carry precomputed paired-bootstrap intervals; the eight extension seeds reproduce the same pattern (seed-level 95% PI $[-1.4, +0.3]$).

Cross-metric consistency across checkpoints.

Table 7 reports the gap on the same fixed output under each of the seven primary evaluators across metrics. Under the original, the gap is positive and large for all decoded-text metrics; teacher-forced NLL is more negative for PURE than GT (-0.419 sign-flipped), consistent with the likelihood-vs-decoded-metric divergence Jiang et al. (2026). Under the six reconstructions the reversal disappears: all chrF and ROUGE-L gaps are negative, all WER gaps favor GT, and sacreBLEU-4 gaps are in $[-0.98, +0.13]$

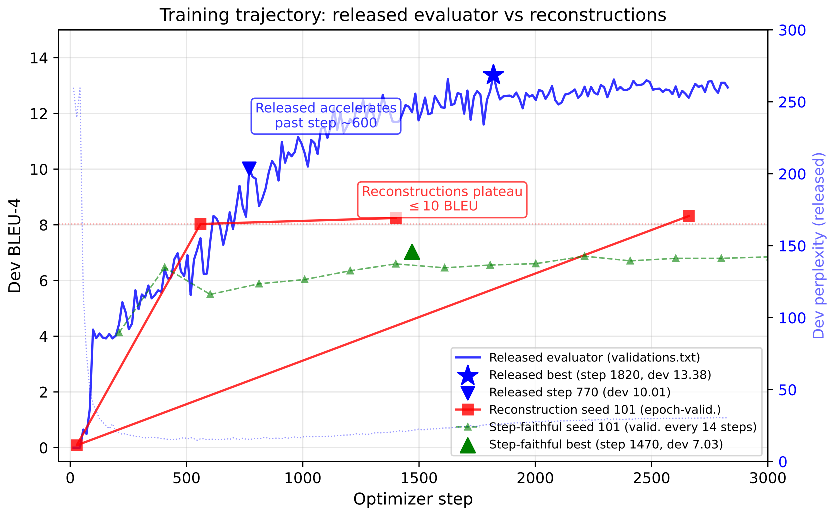

Fig. 2: Training trajectories: released evaluator (blue, decoded-BLEU selection) vs reconstruction seed 101 under epoch-validation (red) and step-faithful protocol (green). The released `best.ckpt` corresponds to step 1,820 (best dev BLEU-4 13.38, blue star). All six primary reconstruction seeds plateau below 8.5 at selection; the released evaluator's dev perplexity bottoms around step 750 and rises thereafter, confirming decoded-BLEU rather than likelihood selection.

with only two small positive cells. The reversal is a property of decoded-text metrics under the original evaluator, not specific to sacreBLEU-4.

Rank stability (see SI Sup. H).

On a fixed nine-system stress-test set, the released evaluator ranks TN-PURE first; five of six reconstructions rank GT first, with moderate cross-checkpoint agreement (mean τ_b=0.44, ρ=0.57).

Reference-validity sensitivity: matched-subset analysis.

Two 2026 corpus audits bear directly on this reversal. Czehmann et al. Czehmann et al. (2026) released manual sign-to-text back-translations of the test set (one deaf fluent L2 DGS signer; per-item confidence flags), documenting information loss, lexical errors, and translationese in the original speech-transcribed references, and showed that reference replacement already changes BLEU scores and rankings on PHOENIX-2014T. Alkain et al. Alkain et al. (2026) quantified train–test text overlap across sign-language translation corpora, showing that a small fraction of highly training-like test items disproportionately inflates corpus BLEU. We reproduce the overlap pattern locally: the 32 test items whose reference exactly matches a training text score GT 40.26 / PURE 78.73 (gap +38.5), while the remaining 609 items score GT 11.93 / PURE 22.20 (gap +10.3).

Table 6: Competence rescue on the released 515-sequence dev split: 20 runs across six hyperparameter families (two to four seeds each; per-seed values in the artifact). All fail the competence gate ($|\Delta$dev BLEU-4$| \leq 1.0$ and $|\Delta$dev WER$| \leq 3.0$; the released evaluator is 13.38 / 83.37; official WER ranges: lr-rescue 87.5–93.1, expanded rescue 86.6–93.1); a long-schedule run reaches a family-maximum 10.02 and still fails. The competence gate conclusion is insensitive to threshold choice: across a $\pm[0.5, 2.0]$ BLEU $\times$ $\pm[1.5, 4.5]$ WER grid of twelve combinations, 0 of 52 trained checkpoints passes any of them.

Variant family	dev BLEU-4 range	min $\lvert\Delta$BLEU-4$\rvert$ vs Original
lr 5×10^{-4} / 2×10^{-3} (8 runs)	7.9–9.7	3.7
batch size 128 / 512	6.8–9.2	4.2
dropout 0.2	6.9–7.5	5.9
weight decay 0	**9.8–9.9**	**3.5**
label smoothing 0.1	6.8–7.4	6.0
600 epochs, lr 5×10^{-4}	8.2–8.3	5.1

Because Czehmann et al. compared original-reference and human-reference conditions on the same confidence=1 subset (462 of 642 items, excluding low-confidence, partial, or untranslatable items), we follow the same protocol. On the matched 461-item subset (after join with our 641 released test IDs), the original-reference gap is +10.95 BLEU points (GT 14.94 / PURE 25.90), and the human-reference gap is +1.32 (GT 7.35 / PURE 8.66), a paired attenuation of 9.64 BLEU points (88.0%, bootstrap CI $[7.92, 11.49]$; Table 8). The full-641 comparison (+11.01 $\rightarrow$ +1.12, attenuation 89.8%) is reported as sensitivity analysis; it mixes reference replacement with floor effects from low-confidence items and is not the primary estimand. This is a *descriptive contrast of two corpus gaps*, not a causal attribution. Under all six reconstruction checkpoints the gap against human references is negative (-0.15 to -0.65). The asymmetry is consistent across metrics on the matched subset: chrF attenuation 82.2% (+12.73 $\rightarrow$ +2.27), BLEU-1 attenuation 84.0% (+20.01 $\rightarrow$ +3.19). Czehmann et al. also showed BLEURT is less sensitive to reference replacement than BLEU; our BLEURT substitute on the full-641 set gives a gap of +0.165 under original refs and a smaller value under human refs (artifact), consistent with their finding.

A natural concern is that the attenuation is a symmetric floor effect. On the matched subset, switching from original to human references, GT drops 7.59 BLEU-4 points (50.8%) but PURE drops 17.24 points (66.6%) — PURE loses a larger fraction (ratio 1.31×). The asymmetry is even larger under BLEU-1 and chrF, so simple symmetric compression is insufficient, but the asymmetry does not discriminate between reference-text defects and training-distribution alignment.

DGS human evaluation (aborted feasibility study; see SI Sup. J).

We conducted a blind DGS human evaluation with 30 DGS-fluent raters on 100 videos (20 PHX-public target sentences, 5 system outputs per target) as a feasibility probe.

Table 7: PURE–GT gap across metrics and evaluators on PHX-public (641 sequences; canonical decoding) for the original evaluator and the *six primary* reconstructions (of fourteen; the eight extension seeds show the same near-zero/negative pattern in Table 5). chrF, WER and ROUGE-L use the official SLRTP2025 implementations (§4.1). WER and NLL are sign-flipped so positive always means PURE is better than GT. BERTScore-F1 uses `bert-base-multilingual-cased` (bert-score 0.3.12, `lang=de`, `idf=False`, no baseline rescaling); a BLEURT substitute column is in the artifact. The large reversal is consistent in direction across decoded-text metrics under the original evaluator and disappears under all six primary reconstructions; residual reconstruction signs are mixed and near zero for n-gram metrics.

Evaluator	ΔBLEU-4	ΔBLEU-1	ΔchrF	ΔWER†	ΔNLL†	ΔBERTSc.
Original	+11.01	+21.03	+13.23	+13.90	−0.419	+0.066
Seed 101	+0.08	−0.15	−1.31	−2.95	−0.219	−0.005
Seed 202	−0.98	−0.21	−1.94	−3.52	−0.233	−0.006
Seed 303	−0.71	−0.10	−1.20	−1.53	−0.240	+0.001
Seed 404	−0.36	−0.14	−1.20	−1.87	−0.194	−0.001
Seed 505	−0.89	−0.17	−1.78	−3.41	−0.268	−0.004
Seed 606	+0.13	−0.04	−0.38	−3.46	−0.169	−0.001

†WER and NLL are sign-flipped: positive means PURE is better than GT (lower WER, lower NLL).

The UUID-to-system manifest and video files were lost during a host-cleanup operation, so per-system analysis is impossible and the study is treated as an aborted feasibility study, not as audit evidence. The only mapping-independent observation is that raters could distinguish semantic-adequacy levels across the 5 anonymized videos per target (top rank 4.48 vs bottom rank 1.83 on a 1–5 scale); we do not interpret the moderate inter-rater agreement ($\alpha = 0.54$) as evidence of semantic validity, because the lost manifest prevents system attribution. Per-rater response CSVs are preserved in the artifact. Full data and the permutation exploration are in SI Sup. J; a properly-documented re-evaluation with persistent manifest and video hashes would provide the independent semantic criterion needed. Data governance details are in the Ethical Considerations.

Transcript-slot diagnostic (see SI Sup. K).

A rule-based German weather slot extractor computes multiset slot-F1 on the decoded hypotheses. Under the original evaluator, the replay decode attains higher slot-F1 than the reference-pose decode (0.526 vs 0.354), consistent with donor-transcript pass-through; under reconstructions the ordering flips. Extractor validation (50 references + 50 hypotheses, single auditor; P 0.97 / R 0.86) is in SI Sup. K; the extractor observes no spatial grammar or non-manual markers, so the slot diagnostic is not an independent semantic judgment.

Table 8: Matched-subset reference sensitivity (corpus sacreBLEU, BLEU points) under the released evaluator. Primary analysis: confidence=1 subset (461 items) with original and human references computed on the *same* items. Sensitivity analysis: full 641 items. Human references: Czehmann et al. Czehmann et al. (2026) back-translations (CC BY-NC-SA 4.0). Confidence breakdown: 461 confidence=1.0, 121 confidence=0.5, 59 confidence=0.0.

Reference condition	*N*	GT	PURE	Gap
Primary: matched confidence=1 subset				
Original (speech-transcribed)	461	14.94	25.90	+10.95
Human back-translation (conf=1)	461	7.35	8.66	+1.32
Paired attenuation				9.64 BLEU pts (88.0%), CI [7.92, 11.49]
Sensitivity: full 641 items				
Original (speech-transcribed)	641	12.78	23.79	+11.01 [+9.61, +12.43]
Human back-translation (full)	641	5.66	6.79	+1.12 [+0.38, +1.91]
Paired attenuation				9.89 BLEU pts (89.8%)
Human refs, reconstructions (6 seeds)	641	3.49–4.62	3.35–4.35	−0.15 to −0.65

Table 9: Asymmetric floor-effect control (full-641 sensitivity analysis): GT and PURE decode scores under orig
human references. PURE consistently loses a larger absolute and fractional score than GT across all metrics, ma
simple symmetric compression (equal fractional score drop) an insufficient explanation. The matched-461 fractiona
ratio for BLEU-4 is 1.31× (GT 50.8% vs PURE 66.6%), consistent with the full-641 pattern shown here. The asyn
is descriptive; it does not discriminate between reference-text defects and training-distribution alignment.

Metric	GT (orig)	GT (human)	GT drop	PURE (orig)	PURE (human)	PURE
BLEU-4	12.78	5.66	7.11 (55.7%)	23.79	6.79	17.00 (7
BLEU-1	34.40	29.28	5.12 (14.9%)	55.43	32.35	23.08 (4
chrF	34.59	32.75	1.84 (5.3%)	47.82	35.02	12.80 (2
PURE/GT absolute-drop ratio	BLEU-4: 2.39× (17.00/7.11); BLEU-1: 4.51× (23.08/5.12); chrF: 6.97× (12.80/1.84)					
PURE/GT fractional-drop ratio	BLEU-4: 1.28× (71.5%/55.7%); BLEU-1: 2.79× (41.6%/14.9%); chrF: 5.06× (26.8%/5					
Dual-ref gap (orig+human)	BLEU-4: +11.50; BLEU-1: +20.03					

Absolute-drop ratio = (PURE drop in BLEU points) / (GT drop in BLEU points). Fractional-drop ratio = (PURE fractional drop %) / (GT f
drop %). The qualitative conclusion (PURE loses more than GT) is the same under either ratio; the paper's text uses the fractional ratio when d
"larger fraction" claims (1.28× for BLEU-4) and the absolute ratio when describing "2.4× more than GT" statements.

The seven-evaluator ranking matrix confirms the pattern on constructed probes (replay, composition, weak generation, random controls), not on representative competitive SLP systems. Decoder settings are fixed (beam 3, length penalty −1, max length 30; sacreBLEU `tok:13a|smooth:exp|version:2.5.1`); the canonical grid has 60 cells with empty-hypothesis rate zero throughout, and the reversal is expressed through much higher n-gram precision for PURE (P1 60.6 vs 37.0). Evaluator identity, training randomness, optimization trajectory, and competence remain confounded.

Table 10: Donor-origin contrast estimands (sacreBLEU, BLEU points; gaps vs local GT). SEEN = 7,060 training poses; UNSEEN = 640 other test poses. Per-seed cells in the artifact.

Evaluator	SEEN-PURE	UNSEEN-PURE	SEEN−GT	UNSEEN−GT	Estimand
Original	23.79	8.86	+11.01	−3.92	**+14.93**
Seeds (range)	6.83–9.09	6.59–7.98	−1.10 to +0.13	−1.94 to −0.27	−0.15 to +1.22

Template-density and holdout boundary.

Within PHX-public, retrieval exceeded the recorded test poses even in the low-density stratum (+10.7 BLEU points), so template density moderates magnitude rather than gating the reversal; template-disjoint and BT-retrained holdouts and donor-familiarity cross-fitting do not reproduce it (SI Sup. I).

4.3 Donor-origin contrast characterizes the reversal's co-occurrence with training-pool donors

If the released-evaluator reversal requires exposure to training-pool donors, then restricting retrieval to poses the released evaluator has not seen should remove or invert the advantage. Under the original SLRTP2025 evaluator, UNSEEN-PURE-v1 (retrieving from the 640 other test poses) scores 8.86 sacreBLEU versus 12.78 for GT, a gap of −3.92 BLEU points: the +11.01 advantage of SEEN-PURE-v1 does not survive retrieval-pool substitution. The donor-pool substitution estimand (SEEN − GT) − (UNSEEN − GT) = +14.93 BLEU points is large but is a *donor-pool substitution effect*, not a pure exposure effect, because SEEN-PURE-v1 and UNSEEN-PURE-v1 differ simultaneously in pool size (7,060 vs 640), median Jaccard (0.50 vs 0.37), donor reuse (Gini 0.11 vs 0.29), and training-pool exposure. Cross-fit controls reproduce the same direction (CROSSFIT pooled −3.8 BLEU points vs fold-local GT). Under the six reconstructions all pool-origin contrasts lie in $[-0.15, +1.22]$ (Table 10).

Path-dependent decomposition and balanced factorial.

We decompose the substitution estimand descriptively along telescoping paths (SI Sup. E). The path-based origin estimate varies from +5.8 to +8.4 BLEU across admissible orderings because size and selection interact. The better-balanced 605-query factorial (matching seen/unseen donors at high/low LaBSE similarity within predeclared tolerances) estimates the pool-origin main effect at +2.08 $[+1.56, +2.57]$, the similarity main effect at +6.96 $[+5.93, +8.08]$, and the seen×similarity interaction at +3.70 $[+2.63, +4.79]$ (all Holm p=0.0003). Across reconstructions, the seen main effect shrinks to +0.51 and the interaction is null (−0.39). Two conclusions: Jaccard-only paths inflate the pool-origin component, and the released evaluator's seen advantage *grows with donor–query similarity* while reconstruction interactions are null. The caveat: seen donors remain more similar than unseen within strata (SMD +0.4), so this is a train-pool vs test-pool provenance contrast, not an identified training-exposure effect.

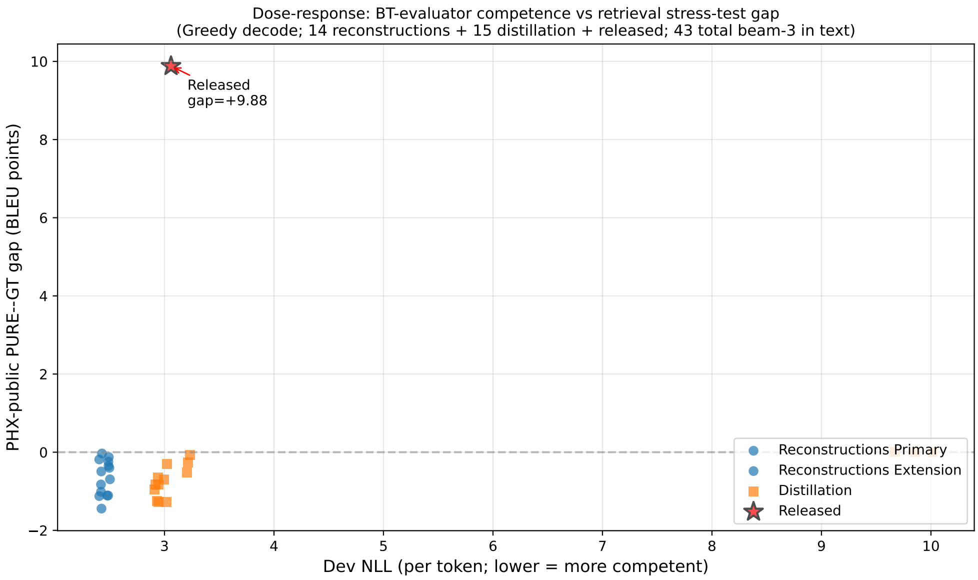

Fig. 3: PURE–GT gap versus dev competence (greedy decode; 14 reconstruction seeds + 15 distillation students + released evaluator = 30 plotted points). The x-axis is teacher-forced dev NLL (lower = more competent). An additional 13 checkpoints (ladder, config-faithful, step-faithful, confirmation, long-schedule, wd0 rescue) have beam-3 PHX-public gaps in the same range ($[-0.71, +0.25]$) but use a different dev-metric protocol and are not plotted; their values are in SI Sup. D. All 43 beam-3 evaluated checkpoints have gap $\leq +0.25$. The figure is *exploratory*; the 43 points are NOT IID replicates. The released evaluator (star) sits far above the constructible family cloud at dev BLEU-4 13.38, gap +11.01. No synthetic or approximate points are used.

4.4 What distinguishes the released evaluator: five converging diagnostic views

We now turn the non-reproduction result into a positive characterization through five related diagnostic views (Fig. 3). These views are *not* independent confirmations: they reuse the same released checkpoint, PHX-public data, and related donor constructions, and are best read as triangulation on one signature. Table 11 summarizes the views (full version in SI Sup. D); the text gives the essential numbers.

Readout* with *generalization: the distinctive combination.

The signature spans two conditions that usually trade off. The released evaluator decodes its full 7,060-item training pool at free-decode BLEU 78.8 with 70.7% exact-match and teacher-forced top-1 accuracy 96.5% — strong readout — while retaining non-trivial in-domain dev/test performance: dev/test free BLEU is 13.38/12.78 with ≈3% exact match. The family separates these regimes: validation-selected checkpoints

Table 11: Summary of the five diagnostic views. The released evaluator sits far outside the family range on all five views. Per-view family composition in SI Sup. D.

Diagnostic view	Family n	Released evaluator	Family range	Gap to family max
(i) Gap–competence	43	+11.01 at dev 13.38	$[-2.01, +0.25]$ at dev 4.5–10.9	+10.76
(ii) Pass-through ratio	24	3.09 (sentence)	1.05–1.31	+1.78
(iii) Train-pool readout BLEU	3	78.8 (70.7% EM)	9.2–11.5 (∼2% EM)	+67.3
(iv) Frame-reversal loss	1	−16.1 (PURE)	−4.3 (wd0 rescue)	|11.8|
(v) Competence gate	52	13.38	4.5–10.02 (max)	+3.36

show train ≈ dev BLEU (11.5 ≈ 9.9), while overfit rescue endpoints reach 99.9 train BLEU but collapse to dev NLL 4.90. Its donor-content pass-through is a 2.4× outlier: the released evaluator decodes donor poses against the donor's *own transcript* at mean sentence-BLEU 0.795 vs 0.257 against the query (ratio 3.09, 95% CI $[2.89, 3.33]$); the family stays at 1.05–1.31.

Leakage sanity checks for the train-pool readout.

Because the 78.8 BLEU / 70.7% EM readout is the most anomalous finding, we ran five checks (full data in SI Sup. N):

1. **Code-path audit.** The `back_translate()` function calls `model.run_batch(translation_beam_size=3)` which invokes beam search; the decoder input is constructed from `<bos>` plus the model's own previously generated tokens only. The reference text enters the pipeline *after* decoding for BLEU scoring, never as a model input. This excludes direct target-token teacher forcing but does not by itself exclude ID-based or cache-based side channels.
2. **Label permutation (output–reference alignment).** Shuffling reference labels across the first 1,000 decoded training items collapses corpus BLEU from 78.6 to 1.06 / 1.30 / 0.90 (three permutations) and EM to ≈0%. This confirms that the decoded output is aligned with its paired reference; however, because the permutation is applied *after* decoding, this test alone cannot distinguish pose-conditioned output from ID-based or cache-based retrieval. A proper input-side test (shuffling pose tensors before decoding while holding IDs fixed) is needed to fully exclude those channels and is reported below.
3. **Input-side pose permutation with output-equivariance verification.** We held item IDs and text metadata fixed, shuffled the *pose tensors* across 100 training items *before decoding*, and verified output equivariance: $E(\text{pose}_j, \text{slot}_i) = E(\text{pose}_j, \text{slot}_j)$ for 100/100 items (exact string match), where "slot" denotes the batch position and metadata context of item i. The permuted decode scored against slot-i references gives BLEU 0.95; scored against the donor (j) references gives BLEU 75.62, identical to the original decode. Decoded-token output length follows

the pose tensor, not the slot (100% donor-length match, 5% slot-length match). Three degenerate-pose controls (zero, noise, repeated pose) also collapse BLEU to <1. This rules out a pose-ignorant pure ID/cache lookup under the tested decoding path and confirms sensitivity to the substituted pose tensor; it does not by itself identify the training mechanism.

4. **Stratified EM.** Exact-match rate is high across all 9 signers (59.0–74.8%), all length bins (short $\leq$5 words: 83.7%; very long >20: 68.6%), and even unique training texts that appear only once (70.2% EM, n=6,777), not just highly repeated templates (94.0%, n=150). Of the 7,060 decoded hypotheses, 74.8% match some training text exactly, and 70% match exactly one — their correct paired reference.
5. **Same-signer held-out control.** Decoding 500 test-split poses (unseen in training) from the *same* 9 signers gives BLEU 12.19 / EM 3.2% vs. train BLEU 78.8 / EM 70.7%. The 67-point BLEU gap between seen and unseen items from the same signers is consistent with training-pool exposure producing the readout. This is a same-signer held-out control, not a fully matched membership control: the test items are not matched on length, text-similarity, broadcast date, or template density, so the gap reflects a combination of training exposure and distributional shift.

The code-path audit excludes direct target-token teacher forcing; the input-side pose permutation with output-equivariance verification rules out a pose-ignorant pure ID/cache lookup under the tested decoding path and confirms sensitivity to the substituted pose tensor. Together with the label permutation, stratified EM, and the same-signer held-out gap, these checks are consistent with the readout reflecting pose-conditioned training-pool exposure. They do not identify the specific training mechanism (which remains undisclosed). The thresholds defining the readout-with-generalization signature were set after observing the released evaluator and are operational heuristics, not validated criteria (§3.3).

Temporal-order sensitivity and constructible-protocol coverage.

The released evaluator loses −16.1 BLEU points when donor frames are reversed (GT loses −5.5, best rescue checkpoint −4.3; full battery in SI Sup. F). Reversal is out-of-distribution, so the differential bounds how much of the score relies on ordered content. None of 52 trained checkpoints across eight constructible-protocol families (reconstructions, rescue, ladder, config-faithful, step-faithful, big-arch, long-schedule, confirmation) passes the competence gate; the config-faithful, big-arch, and long-schedule variants all fail to reproduce the released evaluator's competence, readout signature, or reversal (SI Sup. F).

Knowledge-distillation competence ladder.

To probe the unobserved competence interval above the reconstruction ceiling (dev BLEU-4 10.0), we trained 15 distillation students from the released evaluator as teacher: five distillation strengths ($\alpha \in \{0, 0.25, 0.5, 0.75, 1.0\}$) crossed with three seeds, with loss $\mathcal{L} = (1-\alpha)\,\mathrm{CE} + \alpha\, T^2\, \mathrm{KL}(\mathrm{teacher}\|\mathrm{student})$ at T=2. All 15 completed training and beam-3 PHX-public evaluation. Table 12 shows the distilled students extend the observable range upward (dev BLEU-4 up to 10.9), yet all twelve α<1.0

students have negative PURE–GT gaps (-2.01 to -0.77). The three α=1.0 students produce empty or degenerate outputs (seed 101: 641/641 empty; seed 202: 483/641 empty, 158 single-token "."; seed 303: 641/641 empty), yielding near-zero GT and PURE BLEU and an approximately zero gap. Training-pool readout stays at 9.1–9.9 BLEU / 1.9–2.6% EM versus the released evaluator's readout. Full soft-label distillation (α=1.0) does not transfer the signature either. The readout-with-generalization signature is therefore not a property of the teacher's decoded output distribution that transfers through distillation.

Donor-registry consistency note.

The retrieval algorithm in §3.2 specifies exact-normalized-text exclusion: donors whose NFKC-normalized text equals the query's are skipped. The headline observations (PURE 23.79, gap +11.01) were produced by an earlier frozen materialization that did not apply this exclusion; the 14 non-distillation checkpoints' gap estimates also use this earlier materialization. Applying the §3.2 algorithm (NFKC + exclusion + Levenshtein + SHA-256) lowers the released evaluator's PURE to 23.02 (gap +10.24) and was used to re-decode the 29 trained checkpoints with available model files plus the released evaluator (non-released gap range $[-2.01, 0.00]$; per-item hypotheses in `results/gap_43_canonical_beam3_items/`; `results/gap_43_canonical_beam3.json`). On 610/641 items (95.2%) the §3.2 algorithm produces the same decoded output as the earlier frozen materialization. On the 31 differing items, we decoded each item's top-8 Jaccard candidate poses with the released evaluator and matched the output to the earlier hypothesis: for 30 of 31 items, at least one excluded Jaccard=1.0 candidate (whose NFKC-normalized text is identical to the query) produced a matching decode; one item remained unresolved (`results/donor_identification_31.json`). This is consistent with the exclusion rule explaining most output differences, but does not uniquely recover the historical donor registry. A 14-reconstruction beam-3 sensitivity comparison shows mean $|\Delta\text{gap}|$ = 0.65 BLEU (max 1.45); all non-released gaps remain far below the released evaluator's reversal. Tie-candidate analysis for a subset of differing items is in `results/diff16_tie_analysis.json`. The distillation gaps in Table 12 use the §3.2 algorithm.

Competence extrapolation (exploratory).

Three fits (OLS linear, OLS quadratic, GP with RBF + white kernel) over the 43 checkpoints spanning dev BLEU-4 $[4.5, 10.9]$ all predict negative gaps at 13.38, with substantial model sensitivity (OLS quadratic flips from -13.5 to $+1.3$ when distillation is removed; GP upper bound stays below +2.0 across leave-one-family-out folds). We treat this as an exploratory visual guide (full fit table in SI Sup. L), not as evidence that competence alone cannot explain the reversal. The unobserved 10.9–13.4 interval could in principle harbor non-linear behavior.

Synthesis.

Across the five diagnostic views, the released evaluator is the only checkpoint combining high training-pool readout (78.8 BLEU, 70.7% EM) with retained in-domain

Table 12: Per-α distillation ladder summary (15/15 beam-3 evaluated with canonical donor registry from §3.2, 15/15 with dev BLEU-4). α=0 = hard-label only; α=1 = full soft-label KL. All α<1.0 yield negative gaps; α=1.0 produces empty/degenerate outputs (641/641 or 483/641 empty hypotheses). See §4.4 for registry materialization consistency. Student train readout on first 1,000 training items for 12 students and on full 7,060 for three α=1.0 students; released evaluator readout on full 7,060 for direct comparison.

α	n	dev BLEU-4 (max)	gap (range)	train readout (BLEU / EM%)	reaches 10.5?
0.0 (hard-label)	3	10.51	$[-1.71, -0.80]$	9.1–9.8 / 2.1–2.6%	1/3
0.25	3	10.75	$[-2.01, -0.77]$	9.4–9.5 / 2.0–2.1%	2/3
0.5	3	10.41	$[-1.38, -0.77]$	9.2–9.9 / 2.1%	0/3
0.75	3	10.93	$[-1.21, -0.81]$	9.7 / 1.9–2.2%	2/3
1.0 (full KL)	3	10.81	$[0.00, 0.00]$	9.2–9.7 / 1.9%	1/3
Released (full 7,060, beam=3)	1	13.38	+10.24	78.82 / 70.7%	—

performance (13.38/12.78 dev/test). No reconstruction or distilled student reproduces either the +11.01 gap or this readout-with-generalization signature.

Step-faithful protocol correction.

Two seeds under the released configuration's corrected step-based validation (every 14 optimizer steps) reached dev BLEU 7.03 and 7.47 with near-zero gaps (+0.02, −0.30). This yields *lower* dev BLEU than the epoch-validation approximation (7.0–7.5 vs 8.6–9.5), documenting that validation frequency alone shifts the final checkpoint by 1.5–2.0 BLEU points.

Frozen-protocol confirmation.

Two fresh seeds (1506, 1607), trained after the diagnostic protocol was frozen, confirm the family pattern (gaps −1.03, −0.98; pass-through 1.22, 1.29).

4.5 Checkpoint-targeted oracle stress tests (SI-diagnosed)

We treat evaluator-aware donor selection as a checkpoint-targeted oracle stress test, not as a deployable SLP method, and we report its unmasked advantage as a non-robust diagnostic: the diagonal advantage does not survive reference-overlap exclusion (SI Sup. H). The full cross-checkpoint transfer matrix, candidate-pool sweep, and selector/evaluator-label permutation test are likewise in SI Sup. H because they share target references and evaluator by design. The main text states only that the unmasked advantage is not robust to reference-overlap exclusion and that the white-box search budget (candidate-pool size) controls whether any decoded-BLEU margin is detectable.

5 Discussion

The audit yields a broad negative result (no constructible reconstruction or distilled student reproduces a gap remotely comparable to +11.01 across 43 beam-3 gap estimates (29 of the 43 trained checkpoints re-decoded with the canonical donor registry from §3.2, plus the released evaluator re-decoded separately; 14 retain prior-materialization aggregates; 95.2% decoded-output agreement between materializations; §4.4) (dev BLEU-4 4.5–10.9; six small positive estimates are $\leq +0.25$)) and a descriptive positive characterization: the released evaluator carries a high training-pool readout combined with retained in-domain performance that no public artifact in our family exhibits (§4.4). The 14 same-recipe reconstruction seeds (shared architecture, tokenizer, manifest, selection rule; seeds 101–1405) yield a Student-t prediction interval of $[-1.42, +0.32]$ BLEU points for the PURE–GT gap, while the heterogeneous families (distillation, ladder, config-faithful, step-faithful, rescue, confirmation) are reported separately as exploratory dose-response data, not as IID replicates of a single training process (SI Sup. D). Because all constructible checkpoints have lower competence than the released evaluator (family maximum dev BLEU-4 10.9 vs. released 13.38), this non-reproduction is observed under a competence confound — we cannot rule out that a nonlinear phenomenon emerges in the unobserved 10.9–13.4 interval; an exploratory OLS/GP extrapolation is reported in SI Sup. L for visualization only, without inferential claims. Because Saunders et al. Saunders et al. (2021) recommended BT for comparing between similar model configurations, and our evaluator family is additionally competence-unmatched (all checkpoints dev BLEU-4 < 10.9 vs. released 13.38), our non-reproduction result is scoped to the public training recipe and does not constitute a strong test of their rank-stability proposition.

On the matched 461-item confidence=1 subset, human-reference substitution descriptively attenuates the gap by 88.0% ($+10.95 \rightarrow +1.32$, paired bootstrap CI for attenuation $[7.92, 11.49]$); the asymmetric PURE-vs-GT drop ratio (1.31$\times$ fractional for BLEU-4) is inconsistent with a simple symmetric compression model but does not discriminate between reference-text defects and training-distribution alignment as non-exclusive candidate mechanisms. Czehmann et al. Czehmann et al. (2026) already showed that reference replacement changes BLEU scores and rankings on PHOENIX-2014T; our contribution on this axis is applying the matched-subset protocol to the released SLRTP2025 evaluator and quantifying paired attenuation, not establishing reference-set sensitivity as a novel phenomenon. The headline gap is robust to cluster-aware resampling: signer (9 clusters), broadcast-show (2 clusters: *heute*/*tagesschau*), broadcast-date (297 clusters), and signer$\times$show (17 clusters) bootstrap CIs are all consistent with the query-level CI, though the small cluster counts (especially show with only 2 clusters) make these exploratory sensitivity checks rather than confirmatory tests (SI Sup. G).

Relation to prior work.

These findings extend prior critiques of BT validity from output faithfulness to ranking stability: improved BT scores need not imply target conditioning Hong and Košecká, Jana (2026), low recorded-pose BT baselines are documented Walsh et al. (2024), and BT likelihood is a more stable correlate of human judgment than decoded BLEU Jiang et al. (2026). Our reversal is a *decoded-sacreBLEU* reversal; teacher-forced BT likelihood does not reproduce it (GT NLL 3.060 vs PURE 3.479), consistent with that divergence. Whole-donor replay is a constructed probe, whereas SOKE-style word-level retrieval can improve pose quality under a different representation Zuo et al. (2025); and multiple text scorers on the same BT decoding are not independent confirmation.

Validation frequency is an underappreciated hyperparameter.

The step-faithful vs epoch-validation discrepancy (§4.4) shows that validation frequency itself shapes the final checkpoint: validating every 14 steps rather than every 14 epochs yields *lower* best dev BLEU-4 (7.0–7.5 vs 8.6–9.5), a 1.5–2.0 BLEU-point swing from a single rarely-reported hyperparameter that may contribute to the gap between released and reconstructed evaluators.

Independent reproduction with greedy decoding.

We reproduced the training and evaluation pipeline using greedy decoding (beam size 1). Under greedy decode, the released evaluator scores GT 12.24 and PURE 22.12 (gap +9.88), and no trained checkpoint reproduces the reversal (14 reconstructions: $[-1.44, -0.03]$; 15 distillation students: $\alpha<1.0$ in $[-1.28, -0.08]$, $\alpha=1.0$ near-zero (empty/degenerate outputs: 641/641 or 483/641 empty hypotheses)). The qualitative conclusions are identical to beam-3. Greedy-decode training logs and decoded cells are in the artifact at `results/cells_greedy/` (see Code availability).

What would be needed to identify the mechanism.

Identifying the specific mechanism would require the SLRTP2025 organizers to disclose (i) the exact training command and seeds, (ii) the data pipeline beyond the documented [::2] subsampling, (iii) any curriculum or checkpoint averaging, and (iv) intermediate checkpoints. A *leave-one-donor-out* retrain sweep is computationally tractable (∼1,400 GPU-hours) but requires the original training infrastructure, which is not publicly available.

Actionable recommendations for benchmark organizers.

These recommendations are scoped to the constructed replay-probe stress case on PHX-public, not to general SLP leaderboard stability.

- Release the training command, seeds, data pipeline scripts, and a SHA-256 manifest of training tensors (beyond the released `config.yaml`).
- Release intermediate checkpoints (at least every 100 epochs) so trajectory-level analyses can localize when distinctive properties emerge.
- Leaderboard entries that retrieve from the training pool should attach $\tau \in \{0.3, 0.5\}$ donor-exclusion results as a shortcut-robustness certificate.
- Adopt Czehmann et al. Czehmann et al. (2026) human back-translations as a secondary reference set and follow their matched-subset protocol (confidence=1 items) when comparing original-reference and human-reference conditions; flag systems whose gap diverges sharply between the two.
- Report training-pool free-decode BLEU ÷ dev free-decode BLEU (released: 6.1; reconstructions: 1.1–1.4) as a descriptive metric. Run the input-side pose permutation with fixed IDs and output-equivariance verification (SI Sup. N) to rule out a pose-ignorant pure ID/cache lookup.

6 Limitations

The primary evidence is confined to PHX-public, a template-dense weather domain with documented reference-validity and train–test-overlap issues Czehmann et al. (2026); Alkain et al. (2026). The fixed nine-system stress-test set comprises replay, composition, weak generation, and random controls rather than competitive SLP systems, so the audit demonstrates a stress case on a released evaluator, not a general leaderboard-rank instability. The readout-with-generalization signature has been verified only on the released PHX evaluator; CSL-Daily (SI Sup. M) provides a weak boundary control (train≈test BLEU ratio ∼0.9 versus PHX's 6.1) but its recognizer is too limited for strong cross-dataset validation. We have not verified the readout ratio on other publicly released BT evaluators (e.g., How2Sign, WMT-SL), so the claim that the signature is unique to the released PHX checkpoint should be read as scoped to PHX-public rather than universal.

All 67 non-holdout checkpoints with dev metrics are less competent than the released evaluator (family reaches dev BLEU-4 4.5–10.9, none reaches 13.38). This is the core competence confound: the original training history is unpublished, so competence-matched replication was not possible, and none of the 52 gate-eligible checkpoints passes the prespecified competence gate ($|\Delta\text{dev BLEU-4}| \leq 1.0$ and $|\Delta\text{dev WER}| \leq 3.0$). The 14 same-recipe reconstruction seeds yield a prediction interval of $[-1.42, +0.32]$, valid for the reconstruction recipe but not extrapolable to the released evaluator; the 29 heterogeneous checkpoints (distillation, ladder, config-faithful, step-faithful, rescue, confirmation) are reported as exploratory dose-response data and are not IID replicates of a single training process. An exploratory OLS/GP extrapolation over the 43 checkpoints (SI Sup. L) predicts negative gaps at 13.38 with substantial model sensitivity (OLS quadratic flips sign when distillation is removed); we report these fits only as descriptive visualization. The unobserved 10.9–13.4 interval could in principle harbor non-linear behavior that the present experiments cannot rule out. Saunders et al. Saunders et al. (2021) limited their rank-stability claim to similar model configurations; the large competence gap means our result does not directly test their claim under the conditions they specified. The specific mechanism producing the released evaluator's readout-with-generalization property is not identifiable from public artifacts: neither recipe-conditioned reconstruction, rescue sweeps, config-faithful retraining, nor knowledge distillation (any α, including full soft-label KL) transfers the property.

The donor-pool substitution decomposition is path-dependent (SI Sup. E); pool-origin contrasts are descriptive associations, not identified training-exposure effects. The asymmetric floor-effect analysis (§4.2.1) is a descriptive asymmetry, not a formal test discriminating between reference-text defects and training-distribution alignment; both are non-exclusive candidate mechanisms. The human back-translations are a single-annotator derivative resource whose high-confidence subset lacks second-annotator adjudication. The reference-validity analysis follows Czehmann et al.'s matched-subset protocol; the 88.0% attenuation on the 461-item confidence=1 subset is consistent with their finding that reference replacement changes PHOENIX-2014T scores.

The DGS human evaluation is reported as an aborted feasibility study (data governance in Ethical Considerations); it does not enter the main empirical claims.

The reversal is a *decoded-BT-text-metric* reversal, not a sign-language quality reversal. Without target-conditioned DGS signer judgment or a validated sign-native pose metric, we cannot determine whether the retrieved donor motion conveys target-equivalent meaning; we explicitly do not claim that donor motion fails to express the target, nor that the reversal indicates dataset contamination or pose memorization. The SiLVERScore implementation accepts RGB video or I3D features, not pose tensors, and the SignCLIP-style 203-landmark input cannot be unambiguously mapped from the released 178-point skeleton, so we do not run a sign-native metric under an unvalidated mapping Imai (2025); Jiang et al. (2026).

For completeness, BLEURT scores with `Elron/bleurt-large-512` agree in direction with the other decoded-text metrics (gap +0.165 under the original); rank correlation with canonical BLEURT-20 is $r = 0.87$ but absolute scores are biased (-0.47), so we report them in the artifact only.

7 Conclusion

This audit documents three findings about the released SLRTP2025 back-translation evaluator on a single PHX-public constructed replay-probe case. First, its +11.01 BLEU-point reversal on a constructed whole-donor replay probe does not reproduce at comparable magnitude across 43 beam-3 gap estimates (29 of the 43 trained checkpoints re-decoded with the canonical donor registry from §3.2, plus the released evaluator re-decoded separately; 14 retain prior-materialization aggregates; 95.2% decoded-output agreement between materializations; §4.4) (range $[-2.01, +0.25]$ at dev BLEU-4 4.5–10.9; 14 same-recipe seeds PI $[-1.42, +0.32]$); this non-reproduction is observed under a competence confound, since all 67 non-holdout checkpoints are less competent than the released evaluator, and is scoped to the public training recipe rather than constituting a counter-disproof of Saunders et al.'s rank-stability claim, which was made for similar model configurations. Second, on the matched 461-item confidence=1 subset, human-reference substitution descriptively attenuates the gap by 88.0% (+10.95 $\rightarrow$ +1.32, paired bootstrap CI $[7.92, 11.49]$), consistent with Czehmann et al.'s prior finding that reference replacement changes PHOENIX-2014T scores, but not a causal attribution. Third, the released evaluator exhibits a post-hoc operational signature — high training-pool readout (78.8 BLEU, 70.7% exact match) with retained in-domain dev/test performance (13.38/12.78) — that no constructible reconstruction or distilled student exhibits; pose–text permutation, input-side pose permutation with output-equivariance verification, stratified EM, and same-signer held-out controls rule out a pose-ignorant pure ID/cache lookup and are consistent with pose-conditioned training-pool exposure. The reversal is confined to decoded BT-text metrics; teacher-forced BT likelihood does not reproduce it. These properties hold as far as observable on PHX-public; the readout signature has not been verified on other released BT evaluators, and the fixed nine-system stress-test set comprises replay, composition, weak generation, and random controls rather than competitive SLP systems. The claim is not an evaluator-identity claim, not a leaderboard-rank statement, and not evidence that a randomly reseeded reconstruction would behave alike.

8 Ethical Considerations

Audit design.

No Deaf community member was involved in the audit design or probe construction. The probe (TN-PURE-v1) optimizes for a text-level stress signal rather than sign-linguistic naturalness. The slot extractor's vocabulary was authored by hearing team members; future work should partner with Deaf DGS reviewers. The reference-validity decomposition relies on single-annotator human back-translations of Czehmann et al. Czehmann et al. (2026). Deaf-led reviews identify the absence of Deaf participation as a structural risk in sign-language AI Desai et al. (2024).

DGS human evaluation and data governance.

A blind DGS human evaluation with 30 DGS-fluent raters (100 videos, 20 target sentences) was conducted as a feasibility study (SI Sup. J). The protocol was reviewed under a human-subjects exemption by the authors' institutional ethics committee (`[INSTITUTION-BLINDED] / [IRB-PROTOCOL-ID-BLINDED]`, restored in the camera-ready version). Raters were recruited through DGS-fluent channels and provided informed consent; only aggregate statistics (means, SDs, Krippendorff's α, rank-based profiles) are reported. The video files and UUID-to-system manifest were lost during a host-cleanup operation (2026-07); the loss was discovered during artifact preparation and documented internally. Per-rater response CSVs are preserved at `artifact/results/human_eval/`. Because the loss prevents per-system analysis, the study is reported as an aborted feasibility study, not as audit evidence. The data retention policy for future human evaluations specifies: (i) persistent SHA-256 hashing of all video stimuli at collection time, (ii) redundant storage of the UUID-to-system manifest on independent media, and (iii) a 5-year minimum retention for all de-identified response data. Whether the loss was reported to the ethics review body is being assessed; if required by the institution's policy, a data-incident report will be filed.

Use of public data.

All experiments use publicly released datasets (PHOENIX-2014T, CSL-Daily, SLRTP2025) and Czehmann et al.'s human back-translations. Pose tensors may retain signer identity; we release no per-signer raw poses.

Supplementary Information (SI)

The following supplementary materials, referenced throughout the main text, are available as a separate file (`supplementary.pdf`) and in the anonymous artifact:

- Sup. A: Missing Test ID Sensitivity (`sec:appendix_missing_id`).
- Sup. B: Per-seed Extension Cells 707–1405.
- Sup. C: Analysis Provenance and Multiplicity (`sec:provenance`).

- Sup. D: Canonical Checkpoint Registry details (`tab:checkpoint_registry`).
- Sup. E: Donor-Pool Substitution Decomposition (`sec:appendix_decomposition`, `tab:path_sensitivity`).
- Sup. F: Config-Faithful and Perturbation Details (`sec:appendix_config_faithful`, `sec:appendix_perturbation`, `tab:perturbation`).
- Sup. G: Cluster-Aware Bootstrap (`sec:appendix_cluster_bootstrap`).
- Sup. H: Supplementary Oracle Diagnostics (`sec:appendix_oracle`).
- Sup. I: Template-Density, Holdout Boundary, Donor-Exclusion (`sec:appendix_template_boundary`).
- Sup. J: DGS Human Evaluation Details (`sec:appendix_human_eval`).
- Sup. K: Transcript-Slot Diagnostic Details (`sec:appendix_slot`).
- Sup. L: Competence Extrapolation Details (`sec:appendix_extrapolation`).
- Sup. M: CSL-Daily Boundary Control (`sec:csl-daily`).
- Sup. N: Leakage Sanity Checks for Train-Pool Readout (`sec:appendix_leakage`).

SI availability.

All numerical values, plots, per-checkpoint data, and the greedy-decode reproduction are also available in machine-readable form at https://anonymous.4open.science/r/rcp-audit-artifact-B314/README.md.

Checkpoint Registry (summary)

Declarations

Funding.

Anonymous.

Conflicts of interest.

The authors declare no conflicts of interest.

Ethics statement.

All experiments use publicly released datasets (PHOENIX-2014T, CSL-Daily, SLRTP2025) and Czehmann et al.'s publicly released human back-translations. For the DGS human evaluation (SI Sup. J), 30 DGS-fluent raters rated 100 anonymised videos under informed consent. The protocol was approved under a human-subjects exemption by the authors' institutional ethics committee (reviewing body and reference number: `[INSTITUTION-BLINDED] / [IRB-PROTOCOL-ID-BLINDED]`, restored in the camera-ready version). No PII is collected; only aggregate statistics (mean, SD, Krippendorff's α, rank-based profiles) are reported. The single auditor of the rule-based German weather slot extractor (50 references + 50 hypotheses) was one of the authors. The video files and UUID-to-system manifest were lost during a host-cleanup operation; per-rater response CSVs remain available in the anonymous artifact at `artifact/results/human_eval/`.

Table 13: Canonical checkpoint registry. 70 planned training runs, all 70 completed training and beam-3 PHX-public evaluation. 67 non-holdout runs have dev metrics; 52 of these are non-distillation and enter the competence gate. 43 runs have both dev BLEU-4 and PHX-public PURE–GT gap (extrapolation-eligible). Plus the released original. Each family row's columns sum to the Total row. Full per-checkpoint details in SI Sup. D.

Family	Planned	Trained	Has gap (beam-3)	Has dev	Enters
Reconstructions (primary)	6	6	6	6	multiseed, bootstrap, dose–resp.
Reconstructions (extension)	8	8	8	8	prediction interval, dose–resp.
Rescue lr (A/B × 4)	8	8	0	8	gate counts only
Rescue expanded	12	12	1	12	gate; wd0 also dose–resp., perturb.
Train-pool ladder	4	4	4	4	dose–resp., gap ranges
Config-faithful	4	4	4	4	dose–resp.
Step-faithful	2	2	2	2	dose–resp., validation-frequency
Large-arch (A1–A4)	4	4	0	4	competence attempt
Confirmation	2	2	2	2	dose–resp., family-side confirm.
Long-schedule	2	2	1	2	gate; best seed dose–resp. (dev 10.02)
BT-retrained	1	1	0	0	holdout series
Cross-fit A/B	2	2	0	0	holdout series
Distillation students	15	15	15	15	ladder, dose–resp.
Total	70	70	43	67	
Released original	—	—	1	1	audit target (not trained)

Data licensing and availability.

PHOENIX-2014T is distributed by RWTH Aachen under CC BY-NC-SA 3.0 (checked 2026-06-10). Human back-translations of Czehmann et al. Czehmann et al. (2026) are a derivative resource under CC BY-NC-SA 4.0 covering only their annotations. CSL-Daily and SLRTP2025 are obtained under provider-specific terms. We do not redistribute source pose/video data; the artifact contains only sequence IDs, manifests, hashes, code, and derived statistics. Pose tensors may retain signer identity; we release no per-signer raw poses. Users must obtain PHOENIX-2014T, CSL-Daily, and SLRTP2025 through their official channels.

Code availability.

The anonymous review artifact (https://anonymous.4open.science/r/rcp-audit-artifact-B314/README.md) contains 51 analysis scripts, 60 beam-3 decoded cells (15 evaluators × 4 systems), donor registries, a machine-readable checkpoint registry (71 entries: 70 trained + 1 released), an SHA-256 ledger (396 files), and sufficient statistics for every bootstrap/permutation test. Greedy-decode cells

(30 evaluators $\times$ GT + TN-PURE) are in the artifact at `results/cells_greedy/`. The 30-evaluator canonical beam-3 panel with per-item hypotheses is at `results/gap_43_canonical_beam3_items/` (60 files: 30 $\times$ GT + PURE) plus the donor registry at `results/gap_43_canonical_beam3_items/donor_registry.jsonl`. Run `make core-audit PYTHON=/path/to/python3` for the regression check and 60-cell BLEU decomposition. The artifact ships under an MIT license (`LICENSE`); PHOENIX-2014T, CSL-Daily, and SLRTP2025 datasets remain under their original provider licenses and are not redistributed.

Computational cost.

Experiments ran on one node with eight NVIDIA GeForce RTX 2080 Ti GPUs. Approximate GPU-hours: 1.4 for six primary reconstructions; 1.8 for eight extension seeds; 5.0 for twenty rescue runs; 3.0 for fifteen distillation students; fixed-output decoding 2 minutes per evaluator; 605-query factorial re-scoring 0.17 total. Donor-registry and bootstrap analyses are CPU-bound ($\sim$40 CPU-hours). The two-week wall-clock envelope includes sequential ablations, data preparation, failed protocol checks, and idle queue time.

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